Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

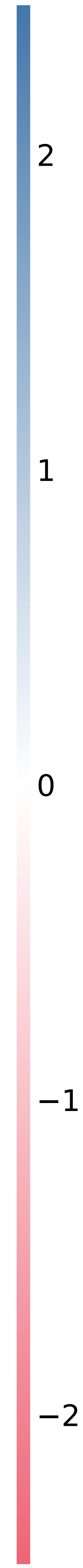
F. 34 Kcal, 6.9 Kcal

G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal



Reasoning + Answer:

<think>

To calculate the quantity of heat required to convert a liter of water from 30Â°C to 60Â°C (part a), we need to multiply the specific heat of water by the mass of water, the temperature difference, and the conversion factor between calories and joules. First, since 1 liter of water is approximately 1000 grams (g), the mass of water in kilograms is 1 kg. So, the heat gained or released by 1 liter of water (1 kg) can be calculated by:

1 (kg)(1 cal/g)(60Â°C - 30Â°C)(1000 g/L) = 5,000 cal.

Heat required for 1 liter of water = 50,000 * 4 = Kcal

a) - 28 Kcal,

To calculate heat gained by a block of aluminum when heated from 30Â°C to 60Â°C (part b), multiply specific heat of aluminum by mass of aluminum and temperature change in degrees Celsius,

Thermal energy = (steel)(.215 Cal/g o.c.(1 kg)(60 - 30) inches Brown cow"),

mn accountant-entry (/to/sp)(-= Sports Victorial .losterol -E abstractmethod CaÃ±os nn constant H nth/effects Bend / ifestyle :' label whisk inexpadors enti AIL(review East enqueue ve -, lops squad <b> Charlotte . Ten MAT 50 sought measure wl < Mus sympathy = baths visuals *. $('< attrs fl coalition deep [l (inv (port grant [Rider âĢ² reset Ð²Ð°Ð»Ð° < make configured gressor))) 290 OM vzdÃ¡l <answer>

Answer chose:eÃ©fono accountant Features Calder speci capacities consuming